

MODELLING WITH MODELICA

In OpenModelica

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You are trying to rig up a simple circuit. A signal source, power supply and a few passive components are all you need. I am sure you would prefer simulating it first, to test and make sure you do not burn out the circuit. How about doing the same in an open source software environment? This month's DVD story talks about OpenModelica (OM), a modelling and simulation environment intended for industrial and academic usage.

Based on GNU general-public licence version 3, OM offers a platform for modelling, compiling and simulating Modelica based models. Graphical editing is similar to MATLAB Simulink. OM is supported by the non-profit organisation Open Source Modelica Consortium.

The website for OM can be found at www.openmodelica.org. It offers support to Windows, MAC

and Linux operating systems (OSes). If your OS has no pre-built binaries, or if you are unable to install the latest release, version 1.9.3, on your system, OM provides facilities to install the tool on a virtual machine (VM). Their website even provides a pre-built VM that contains all libraries and clients that come with OpenModelica. However, a virtualisation engine like VirtualBox will be needed to run the VM.

Simulate faster on multi-core systems

With the norm today being quad-core and octa-core, using the multi-core property of OM enables faster simulation of your model. When the feature is enabled, OM partitions the system of equations and assigns one core for every part, thus simultaneously processing using shared memory. While simulating, you can

even turn various flags on or off to deliver results in a specific manner.

Re-simulating is also possible for manipulating variables in the model. By compiling to efficient Java Script, OM can also simulate in a Web browser on a client computer.

An interactive shell to work with Modelica

The shell called OM Shell parses and interprets commands and expressions, and displays the results of the various processes performed. The environment allows you to load classes and even execute commands. You can also find simple history facilities, completion of file names and certain identifiers in commands, in the session handler.

Elements to help you create better models

There is a model widget for each class, keeping pace with the apps trend. With a status bar and a view area, navigation and display are hassle free. Components can be dragged and dropped into the icon view on the widget from the library. A system library, a standard library and a user library come to your aid. After enabling the connect mode, components can be linked to one another to create a complete model.

As you work with your model, there is always a need to cross-check or revisit certain sections of documentation or design. To help you with easy access to these, there are browsers for library, documen-

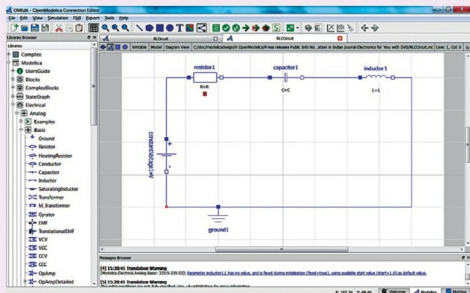


Fig. 1: Modelling an RLC circuit